The 6-Step Data Quality Method: Tasks and Questions

https://github.com/royruddle/6-step-data-quality-method

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Version history

Edition	Date	Description
1	2024-02-06	The original version of the method was published as *A Practical Guide to Characterising Data and Investigating Data Quality* (https://doi.org/10.5518/1481) under the CC BY 4.0 licence
2		Contains 69 tasks (2 renamed; 2 removed; 9 added) and 127 questions (47 revised or new). Tasks renamed: • *Exact duplicates* changed to *Duplicate records* • *Inconsistent duplicates* changed to *Inconsistent values* Tasks removed: • *Characters & symbols* (covered by one of the *Validity* questions) • *Coverage* (subdivided into population, spatial and temporal) Tasks added: • *Overall documentation; Documentation of variables; Documentation of relationships* • *Empty columns; Empty rows; Incorrect column names* • *Population coverage; Spatial coverage; Temporal coverage*

Ruddle, R. A. (2026). The 6-step data quality method. https://github.com/royruddle/6-step-data-quality-method.

Introduction

This data quality method is designed for data scientists to use in their day-to-day work, and is divided into six steps:

1. Is anything obviously wrong (look at your data and any documentation)?
2. Watch out for special values (e.g., indicating a missing or invalid value; you will need to flag them as missing, etc. before proceeding to the next step).
3. Is any data missing (is it safe to proceed or do you need more data?)?
4. Check each variable (is it what you expect?).
5. Check combinations of variables (do they violate any of your assumptions or business rules?).
6. Profile the cleaned data.

The tables below provide a comprehensive list of tasks to perform. The tasks are articulated as questions for you to answer about your data and mapped onto the six steps. Key principles are to avoid rework and nasty surprises late on in your project.

You may approach investigations by establishing the characteristics of your data and/or assessing its quality. The tasks and questions for each of those are described below. We also provide a spreadsheet checklist which lists the tasks that we recommend for each step.

Characterisation tasks & questions

These tasks are divided into groups about whole data sources or files (see Table 1), each variable (see Table 2), the distribution of each variable (see Table 3) and combinations of variables (see Table 4). Within each table, the tasks are listed in alphabetical order and accompanied by one or more questions for you to answer. The answers to these questions can then inform how you might wish to proceed, for example if the range of values falls outside expectations, then you might need to revisit the original source of the data to establish where the error occurred.

ID	Task	Questions to answer about each data source or file	Method Step(s)
C-1	Example values	What does the data look like (print the first/last few rows)?	1, 6
C-2	File format	What text encoding format do the files use?	1
C-3	Number of columns	How many columns are there in the data?	6
C-4	Number of rows	How large is each file (e.g., in gigabytes)? How many rows of data are there?	6

Table 1: Characterisation tasks for each data source or file.

ID	Task	Questions to answer about each variable/field/column	Method Step(s)
C-5	Data format	In what format are the values stored? What are the patterns of the values? Which characters are used in the values?	6
C-6	Data type	What is the data type? Is the analysis software using the correct data type?	6
C-7	Number of infinite values	How many values are flagged as infinite?	2
C-8	Number of null values	How many null values are there?	3
C-9	Number of special values	Which values have a special meaning (e.g., not known, withheld or invalid)? Are there any text values in numerical columns? Are there any unusually small or large integers (e.g., -999 or 999)? How many special values are there?	2
C-10	Number of unique values	How many unique (distinct) values are there? What are those values? Are all the values identical? Does the number of unique values equal the number of records (or the number of values, if some are missing)?	4, 6
C-11	Number of zero values	How many values are equal to zero?	2
C-12	Precision	How many significant figures or decimal places are there?	6
C-13	Units	What are the units?	1
C-14	Value lengths	How long is each value (how many characters are there)? Is every value the same length? If not, what is the minimum/maximum length?	4

Table 2: Characterisation tasks for each variable/field/column.

ID	Task	Questions to answer about variable distributions	Method Step(s)
C-15	First digit	How many times does each digit (0 – 9) appear first?	4
C-16	Frequency measures	What is the distribution of the values? How many times does each value occur? What is the shape of the distribution (e.g., Gaussian, bimodal or skewed)?	4, 6
C-17	Outliers	Are there any outliers?	2, 6
C-18	Range and variability	What is the minimum/maximum value? What is the mean, variance, etc? What are certain percentiles (e.g., 25^{th}, median and 75^{th})?	2, 6

Table 3: Characterisation tasks for distributions of each variable.

ID	Task	Questions to answer about combinations of variable	Method Step(s)
C-19	Clusters	What clusters are there? Are any groups of records similar?	6
C-20	Correlation	What is the correlation coefficient for each pair of numerical variables?	6
C-21	Cross tabulation	How often does each combination of values occur?	6
C-22	Curve fitting	What type of curve (if any) fits each pair of numerical variables?	6
C-23	Primary features	What are the primary features in the data (e.g., principal components)?	6
C-24	Trends	Do any pairs of variables exhibit trends?	6

Table 4: Characterisation tasks for combinations of variable.

Data quality tasks & questions

Data quality is divided in a similar manner to the characterisation tasks. Table 5 is about documentation. Table 6 lists tasks for investigating data quality issues that most often occur when you combine data from different sources or files. Completeness tasks are divided into those that involve individual variables (see Table 7) and combinations of variable (see Table 8). Accuracy tasks Table 9 lists accuracy tasks that involve individual variables, and Table 10 lists tasks for data quality tasks issues that involve combinations of variables.

Data is measured against three kinds of backdrop: time, space and population. Time may be at any granularity (centuries, years, seconds, etc.). Space includes places, geographic regions, other types of area or higher-dimensional spaces. Population may refer to people or any other type of entity (animals, plants, sensors, etc.)

ID	Task	Questions to answer about documentation	Method Step(s)
DQ-1	Overall documentation	Is there any documentation or metadata? Is the documentation/metadata machine-readable?	1
DQ-2	Documentation of variables	Is every variable documented? Is the format specified? Are units specified? Are the allowable values or range specified? Are special values defined (e.g., not known, withheld or invalid)?	1
DQ-3	Documentation of relationships	Are any functional dependencies (business rules) defined? If there are multiple files, is the relationship between them defined?	1

Table 5: Data quality tasks about documentation.

ID	Task	Questions to answer about missing data	Method Step(s)
DQ-4	Empty columns	Are there no values in any columns?	1
DQ-5	Empty rows	Are there no values in any rows?	1
DQ-6	Missing column names	Are any variables missing their name (most often it is the last variable in a table)?	1
DQ-7	Missing header	Are the names of the variables provided (e.g., in the first row)?	1
DQ-8	Missing records	Is the number of records similar to the number you expect? If there are multiple files, do any have notably fewer records (or a much smaller file size) than the others?	1
DQ-9	Missing variables	Does the data contain all of the variables you expected? If there are multiple files for the same data, do they all contain the same variables?	1
	Task	**Questions to answer about duplicate data**	
DQ-10	Duplicate header	Are the variables names repeated?	1
	Task	**Questions to answer about incorrect data**	
DQ-11	Incorrect column names	Are there leading or trailing spaces in any column names? Are there typos in any column names? If there are multiple files for the same data, are the column names the same?	1
	Task	**Questions to answer about inconsistent data**	
DQ-12	Different table structure	Do the files store certain variables in different ways (e.g., date vs. separate day, month and year)?	1
DQ-13	Different word orderings	Do the files store certain variables in different orders (e.g., first name then family name vs. the opposite way around)?	1

Table 6: Data quality tasks for each data source or file.

ID	Task	Questions to answer about missing data	Method Step(s)
DQ-14	Missing values	How many values are missing? Did you expect the variable to be complete?	3, 6
	Task	**Questions to answer about coverage**	
DQ-15	Granularity	Is the data imprecise (e.g., time in hours not minutes) Is the data aggregated too coarsely (e.g., time period, spatial region or population group)? Is the data too fine (what can you discard)?	3, 6
DQ-16	Population coverage	Are you missing data for any part of the population (people, sensor, etc.)? Does the data contain enough samples of each group in the population (e.g., male/female)? Is the data representative of the population (i.e., not biased)?	3, 6
DQ-17	Spatial coverage	Does the data cover the correct areas (e.g., places or geographical regions)? Are there any gaps or holes in the areas? Are any extra areas included (e.g., outside a region)?	3, 6
DQ-18	Temporal coverage	Is the data outdated? Does the data cover the correct time period? Are there gaps in time?	3, 6
	Task	**Questions to answer about duplicates**	
DQ-19	Duplicate records	Are any records identical?	3

Table 7: Completeness tasks for each variable.

ID	Task	Questions to answer about missing data	Method Step(s)
DQ-20	Combinations of missing values	What combinations of variables are missing together? Are there any combinations of variables for which there should always be at least one record?	3
ID	**Task**	**Questions to answer about coverage**	
DQ-21	Inconsistent coverage	Does the data refer to different time periods? Does the data refer to different areas (e.g., places or geographical regions)? Does the data refer to different populations (people, sensor, etc.)?	3
DQ-22	Inconsistent granularity	Does all of the data refer to the same granularity (time, geography, population, etc.)?	3

Table 8: Completeness tasks for combinations of variable.

ID	Task	Questions to answer about duplicates	Method Step(s)
DQ-23	Uniqueness violation	Should every value be unique, and is that true?	4
	Task	**Questions to answer about extreme values**	
DQ-24	Extreme values	Are any values outliers (numerically, dates or times)?	4
DQ-25	Unusual category name	Is the length of any categorical values much shorter/longer than the others? Is any value notably different? Is a value irrelevant for a column?	4
	Task	**Questions to answer about incorrect values**	
DQ-26	Embedded values	Are multiple variables combined into one value (e.g., date and time, or multiple elements of an address)?	4
DQ-27	Misfielded	Do any values appear in the wrong column? Is the same value repeated across columns?	4
DQ-28	Misspelling	Are all words spelt correctly? Is the capitalisation consistent?	4
DQ-29	Naming conflicts	Are there any synonyms or homonyms?	4
DQ-30	Noise	How much variation is there in the values? Does noise mask small values?	4
DQ-31	Validity	Do any values contain characters or symbols that should not be present? Do all of the values exist in the domain (e.g., names of categories or numbers from a specific set of choices)?	4
DQ-32	Wrong data type	Is the data type correct for each variable? Are any dates stored as strings? Are any categorical variables stored as numbers? Is text stored as numbers (e.g., losing leading zeros)? Is numeric data stored as text?	4
DQ-33	Wrong format	Are the values formatted correctly? Are some values decimals but most are integers? Do some values have a different number of decimal places to the other values?	4
DQ-34	Wrong values	Are any values impossible (e.g., dates in the future)? Are any values definitely not correct? Are any values out of range? Should all values be positive but some are negative? Is any text truncated? Were event dates recorded as the date the data were entered into the system?	4
	Task	**Questions to answer about plausible values**	
DQ-35	Implausibly low/high values	What are the minimum/maximum values, and are they plausible? Do any values look suspicious when compared with the other values?	4
DQ-36	Same value for too many records	Do any values occur many times, and is that plausible?	4

Table 9: Accuracy tasks for each variable.

	Task	Questions to answer about inconsistent data	
DQ-37	Different data formats	Are variables of the same type all in the same format?	5
DQ-38	Inconsistent units	Do variables of the same type all use the same units?	5
DQ-39	Inconsistent values	Do any records contradict each other (e.g., different date of birth or ethnicity for a person)?	5
DQ-40	Violate assumptions	What assumptions have you made about variables (are they true)?	5
DQ-41	Violate functional dependencies	Do any sets of values violate rules (e.g., age vs. date of birth)?	5
DQ-42	Violate integrity	Is anything implied by one part of the data missing from another part (e.g., customers but no corresponding account details, or shop open at a given time but no sales data)?	5
	Task	**Questions to answer about plausible values**	
DQ-43	Implausible changes in values	By how much do values change (e.g., person's height from one date to another), and is that plausible?	5
DQ-44	Implausibly low/high combinations of values	If any variables are related (e.g., height and weight), are there any outliers?	5
DQ-45	Implausible range	What is the difference between variables (e.g., start and finish time), and are those differences plausible?	5

Table 10: Data quality tasks for combinations of variable.

